L2: Adam, 500 epochs
Cells: mean ± SE. Bold: within one SE.Square: minimum. Circle: selected.

Hidden layers: (50,)

Sigmoid Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.102 ±0.0038	0.0991 ±0.0053	0.0995 ±0.0051	0.0986 ±0.0049
0.001	0.1 ±0.0042	0.0978 ±0.0048	0.0396 ±0.0045	0.0403 ±0.0045
0.0001	0.0315 ±0.0046	0.023 ±0.004	**0.0195 ±0.005**	**0.0201 ±0.0057**
1e-05	0.0281 ±0.0046	**0.0175 ±0.0031**	**0.0182 ±0.0043**	**0.0187 ±0.0051**

Hidden layers: (100, 100)

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.0994 ±0.0055	0.0995 ±0.0042	0.0988 ±0.0046	0.0991 ±0.0051
0.001	0.0984 ±0.0048	0.0988 ±0.005	0.1 ±0.0051	0.103 ±0.0038
0.0001	0.025 ±0.0033	0.0207 ±0.0065	**0.0188 ±0.0034**	**0.0188 ±0.0033**
1e-05	**0.017 ±0.0032**	**0.0194 ±0.0035**	**0.0186 ±0.0039**	**0.0195 ±0.0028**

Hidden layers: (150,)

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.0987 ±0.0047	0.103 ±0.0046	0.103 ±0.0047	0.108 ±0.0072
0.001	0.0979 ±0.005	0.104 ±0.0066	0.0411 ±0.0045	0.0445 ±0.0025
0.0001	0.0959 ±0.0052	0.0352 ±0.008	0.021 ±0.0038	**0.0194 ±0.003**
1e-05	0.0414 ±0.005	**0.0187 ±0.0037**	**0.018 ±0.0026**	**0.0179 ±0.003**

Hidden layers: (300, 300)

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.105 ±0.0044	0.104 ±0.0056	0.11 ±0.003	0.102 ±0.0052
0.001	0.107 ±0.0082	0.102 ±0.0044	0.101 ±0.0087	0.341 ±0.14
0.0001	0.0283 ±0.0061	**0.0197 ±0.0035**	0.122 ±0.031	0.168 ±0.041
1e-05	**0.0181 ±0.0047**	**0.0189 ±0.0046**	0.0426 ±0.016	0.14 ±0.031

ReLU

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.0845 ±0.0039	0.101 ±0.0038	0.102 ±0.0033	0.101 ±0.0047
0.001	0.0205 ±0.0036	0.0226 ±0.0037	0.0241 ±0.0038	0.0277 ±0.0047
0.0001	0.0219 ±0.0033	**0.0183 ±0.0028**	**0.0189 ±0.0037**	**0.0189 ±0.004**
1e-05	0.0218 ±0.0033	0.0204 ±0.0037	**0.0187 ±0.0037**	**0.0188 ±0.0043**

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.101 ±0.0039	0.101 ±0.0039	0.102 ±0.0034	0.103 ±0.0034
0.001	**0.0176 ±0.0038**	**0.0181 ±0.0035**	0.0201 ±0.0036	0.0203 ±0.0041
0.0001	0.0209 ±0.0025	**0.0196 ±0.0024**	0.0198 ±0.0037	**0.0186 ±0.0034**
1e-05	**0.0196 ±0.0028**	**0.0191 ±0.0021**	0.0201 ±0.0037	**0.0195 ±0.0038**

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.082 ±0.0033	0.101 ±0.0039	0.102 ±0.0034	0.101 ±0.0045
0.001	0.0204 ±0.0037	0.0198 ±0.0033	0.0244 ±0.0037	0.0222 ±0.0035
0.0001	**0.0183 ±0.0033**	**0.0192 ±0.0035**	**0.0192 ±0.0034**	**0.0187 ±0.0039**
1e-05	0.0207 ±0.0034	0.0199 ±0.0039	**0.0162 ±0.0034**	**0.0192 ±0.0039**

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.101 ±0.0037	0.101 ±0.0039	0.101 ±0.0038	0.101 ±0.0039
0.001	**0.0176 ±0.0035**	**0.0175 ±0.0038**	0.0214 ±0.0029	0.0353 ±0.0052
0.0001	0.0211 ±0.0028	0.0199 ±0.0025	0.0202 ±0.0038	0.0295 ±0.0061
1e-05	0.0207 ±0.0027	**0.019 ±0.0028**	0.021 ±0.0033	0.0561 ±0.024

Leaky ReLU

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.0847 ±0.0039	0.101 ±0.0038	0.102 ±0.0033	0.1 ±0.0047
0.001	**0.0206 ±0.0037**	0.0226 ±0.0036	0.0243 ±0.0038	0.0267 ±0.0044
0.0001	0.0218 ±0.0034	**0.0182 ±0.0029**	**0.019 ±0.0034**	**0.0188 ±0.0042**
1e-05	0.0218 ±0.0033	**0.0205 ±0.0038**	**0.0174 ±0.004**	**0.0198 ±0.004**

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.101 ±0.0039	0.101 ±0.0039	0.102 ±0.0035	0.103 ±0.0033
0.001	**0.0181 ±0.0039**	**0.0183 ±0.0039**	**0.0193 ±0.0038**	**0.02 ±0.0039**
0.0001	0.0219 ±0.0019	**0.0206 ±0.0028**	**0.0193 ±0.004**	**0.0196 ±0.0035**
1e-05	**0.0204 ±0.0026**	**0.0213 ±0.0014**	**0.0189 ±0.0026**	**0.0205 ±0.0031**

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.0822 ±0.0033	0.101 ±0.0039	0.102 ±0.0034	0.101 ±0.0045
0.001	**0.0204 ±0.0037**	**0.0199 ±0.0033**	0.0244 ±0.0037	0.0225 ±0.0036
0.0001	**0.0184 ±0.0032**	**0.019 ±0.0035**	**0.02 ±0.0032**	**0.0185 ±0.0039**
1e-05	**0.0205 ±0.0036**	**0.0181 ±0.0039**	**0.019 ±0.0032**	**0.0188 ±0.0041**

Penalty strength λ \ Learning rate γ	0.003	0.01	0.05	0.1
0.1	0.101 ±0.0036	0.101 ±0.0039	0.101 ±0.0037	0.101 ±0.0039
0.001	**0.0177 ±0.0037**	**0.0175 ±0.0037**	**0.0201 ±0.0026**	0.0236 ±0.0034
0.0001	0.0223 ±0.0024	**0.0183 ±0.0025**	**0.0213 ±0.0019**	0.0238 ±0.0027
1e-05	**0.0208 ±0.0025**	**0.0202 ±0.0022**	**0.0206 ±0.0023**	0.0266 ±0.0041

Learning rate γ

Mean CV validation MSE
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}